

Daily Science Anecdotes

Topic: The Birthday Problem: When 23 People Are Enough

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Imagine a room with only 23 people.

Would you expect two of them to share the same birthday?

Most people instinctively say no. There are 365 possible birthdays, after all. Twenty-three people seem like far too few.

But mathematics says something surprisingly different: under the usual assumption of 365 equally likely birthdays, 23 people are enough to make the probability of a shared birthday slightly greater than 50%.

The trick is that we are not asking whether someone shares *your* birthday. We are asking whether **any two people** in the room share a birthday. As the number of people grows, the number of possible pairs grows much faster than our intuition suggests.

For 23 people, there are

$$\binom{23}{2} = \frac{23 \times 22}{2} = 253$$

different pairs of people who could potentially match.

Instead of calculating the probability of at least one match directly, it is easier to calculate the opposite: the probability that **everyone has a different birthday**.

The first person can have any birthday. The second must avoid that one, so the probability is 364/365. The third must avoid two birthdays, giving 363/365, and so on. Therefore,

$$P(\text{no shared birthday}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{343}{365}.$$

So,

$$P(\text{at least one match}) = 1 - P(\text{no match}) \approx 1 - 0.4927 = 0.5073.$$

That is about **50.73%**.

So the surprising result is:

23 people $\Rightarrow$ 50.73% chance of a shared birthday

Why does our intuition struggle with this?

Because we tend to imagine one person asking, “Does anyone have the same birthday as me?” That is a much harder match to find. The birthday problem asks a much broader question: **does anyone match anyone else?**

This is more than a party trick. The same mathematics appears whenever repeated outcomes can occur among many possible pairs: coincidences, duplicate records, and problems in probability and computer science all involve the same underlying idea.

The real lesson is not that 23 is a magical number.

It is that **the number of possible interactions in a group can grow much faster than our intuition expects.**

Sometimes, mathematics surprises us not because the numbers are complicated, but because there are more connections between them than we first notice.